Baryon Asymmetry With CP Violation in SM

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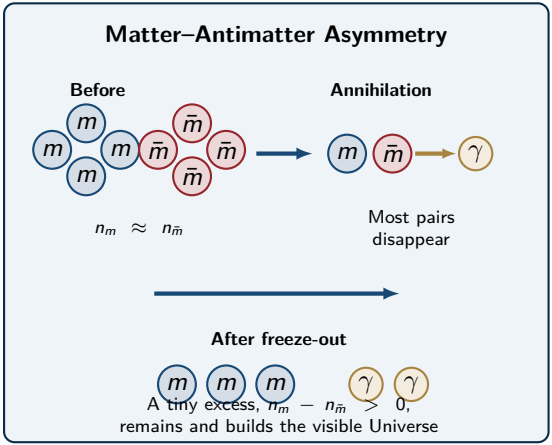


Introduction

Central question

Can the Standard Model, evaluated around the electroweak epoch, generate the observed matter–antimatter asymmetry of the Universe?

- This presentation focuses on electroweak baryogenesis at $T \sim 100\,\mathrm{GeV}$.
- The target observable is

$$\eta_B \equiv \frac{n_B - n_{\bar{B}}}{s} \approx 10^{-10}.$$

- The calculation combines anomalous baryon violation, CKM-induced CP asymmetry, and the finite-temperature effective potential.
- The aim is a quantitative yes/no test of the Standard Model.

3 Sakharov Conditions
$\Delta B \neq 0$, CP violation, non-equilibrium

↓

Electroweak epoch
$T \sim 100\,\mathrm{GeV}$

↓

Effective-action analysis
thermal potential + flavour structure

↓

Quantitative verdict
compare theory with observation

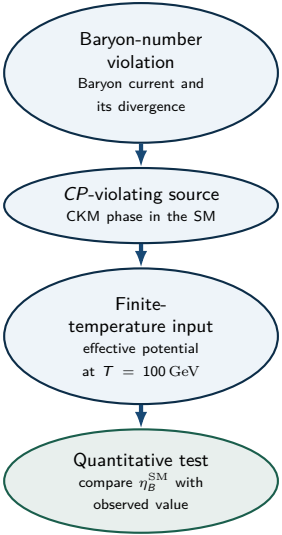
Objectives

What this presentation aims to establish

- Explain how baryon number violation arises in the Standard Model through the CP violation.
- Identify the role of *CP* violation as Sakharov's second condition and connect it to the CKM phase.
- Evaluate the one-loop & finite-temperature effective potential at the electroweak scale and extract the relevant numerical inputs.
- Estimate the baryon-to-entropy ratio η_B within the Standard Model and compare it with the observed asymmetry.

Central question

$$\eta_B^{\mathrm{SM}} \overset{?}{\sim} \eta_B^{\mathrm{obs}} \approx 10^{-10}$$



CP Violation And Baryon Asymmetry

- Under CP transformation, "matter" state into a physical "antimatter" state :
  $CP\,|\text{Particle}_{LH}\rangle \longrightarrow |\text{Antiparticle}_{RH}\rangle, \quad C\,|e^-(\vec{p},s)\rangle = |e^+(\vec{p},s)\rangle$
  $P\,|\psi(\vec{p},s)\rangle = |\psi(-\vec{p},s)\rangle, \quad P\,|\text{helicity}\rangle \rightarrow -|\text{helicity}\rangle$
- If CP symmetry were exact, matter and antimatter would be produced in equal amounts, so no baryon asymmetry would appear.
- In the Standard Model, CP violation comes from the complex phase of the CKM matrix, which controls flavour mixing in weak interactions.
- This CP-violating effect gives a small preference to matter over antimatter, but the effect is too weak to explain the observed asymmetry by itself.

Baryon Current and its Divergence

- The Baryon current j_B^μ and the associated baryon number from the Noether's theorem are related as : $j_B^\mu = \frac{1}{2}\bar{Q}\gamma^\mu Q, \qquad B(t) = \int d^3x\, j_B^0(\mathbf{x}, t)$
- Non-zero divergence of baryon current leads to non-conservation of baryon nummber.

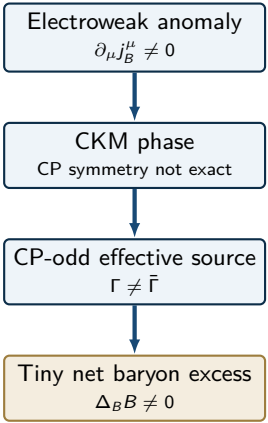
$$\partial_\mu j_B^\mu = n_f \left(\frac{g^2}{32\pi^2} W^a_{\mu\nu} \tilde{W}^{a\,\mu\nu} - \frac{g'^2}{32\pi^2} F_{\mu\nu} \tilde{F}^{\mu\nu} \right),$$

- The expectation value of this term :

$$\langle \partial_\mu j_B^\mu \rangle = \int dA^a_\mu\, d\bar{\Psi}\, d\Psi\, dh\, \partial_\mu j_B^\mu \exp\left[i \int d^4x\, \mathcal{L} \right].$$

CP Violation And Baryon Non-conservation

- **Anomaly:** In the Standard Model, baryon number is not exactly conserved because the electroweak anomaly gives $\partial_\mu j_B^\mu \neq 0$.
- **Source of CP violation:** The CKM matrix contains a complex phase, so a process and its CP-conjugate need not occur with exactly the same amplitude.
- **Effective-action picture:** This phase enters the effective action as a CP-odd contribution, producing a difference between Γ and $\bar{\Gamma}$.
- **Physical consequence:** The difference feeds into $\Delta_B B$, creating a slight preference for baryons over antibaryons during the electroweak era.



CP-Odd Part of the Effective Action

- The net baryon-number change is controlled by the difference between the effective action and its CP conjugate,

$$\eta = \frac{n_B}{n_\gamma},\ n_B = n_b - \bar{n}_b,\ \Delta_B B = \frac{1}{2}(\Delta B - \Delta \bar{B})\ ,\ \eta_1 = \frac{n_B}{s}\ ,\ s \approx g_* n_\gamma\ ,\ \eta_1 = \frac{n_B}{g_* n_\gamma}\ ,\ \boxed{\eta_1 \approx \frac{\Delta_B B}{g_*}}$$

$$\Delta_B B = 12 V t_f\, \delta(0) \left[e^{i\Gamma(T)} - e^{i\bar{\Gamma}(T)} \right].$$

- After CP operation on Lagrangian:

$$e^{i\Gamma} = \int dA^a_\mu\, dh\, d\bar{\Psi}\, d\Psi\ \exp\left[i \int d^4x\, (\mathcal{L}_1 - \bar{d} A d) \right] \exp\left[\int d^4x\, \bar{d} B d \right]$$

$$e^{i\bar{\Gamma}} = \int dA^a_\mu\, dh\, d\bar{\Psi}\, d\Psi \exp\left[i \int d^4x (\mathcal{L}_1 - \bar{d} A d) \right] \exp\left[- \int d^4x\, \bar{d} B d \right]$$

Embedding the CKM Phase into the Effective Action

- The transformed mass matrix is written as

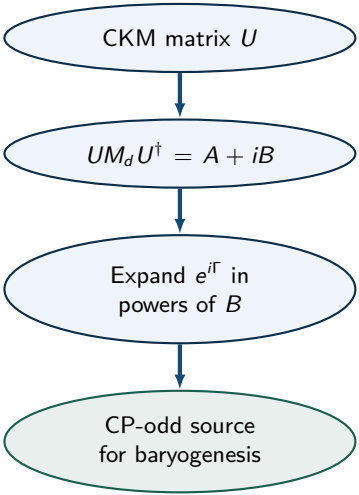
$$UM_d U^\dagger = A + iB.$$

- The relevant Lagrangian becomes
$\mathcal{L} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion kinetic}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}$

$$\mathcal{L} = \mathcal{L}_1 - \bar{d}\,(A + iB)\,d,$$
$$\mathcal{L}(\text{CP}) = \mathcal{L}_1 - \bar{d}\,(A - iB)\,d.$$

- With the effective potential ,the effective action is

$$\Gamma(\Phi) = -V_t f,$$
$$V_{\text{eff}} = -\frac{1}{T^4}\,V_{\text{eff}}.$$



CP Symmetry and Its Breaking

- Therefore the baryon-number change can be written as

$$\Delta_B B = i\,24\,\delta(0)\,\frac{1}{T^4}\sin\left[\frac{U_{ki}B_{ij}U^\dagger_{kj}}{T^4}\left(-\frac{3}{4\pi^2}m'^{\,3}_{d_k}\left(\ln\frac{m'^{\,2}_{d_k}}{\mu^2}-1\right)+\frac{1}{2}m'_{d_k}T^2\right)\right]e^{-\Gamma_{\rm eff}(A)}.$$

- In the Standard Model, the non-zero CKM phase makes *CP* violation possible, but its size remains too small for successful baryogenesis.
- The thermal minimum of the Higgs potential supplies one of the numerical inputs. At the electroweak scale we use

$$V(\Phi_c,T) = D(T^2-T_0^2)\Phi_c^2 - ET\Phi_c^3 + \frac{\lambda(T)}{4}\Phi_c^4, \qquad \frac{\partial V}{\partial \Phi_c} = 0.$$

This gives the working value

$$\Phi_{\rm min}(T\approx 100\,{\rm GeV}) \approx 216.56\,{\rm GeV}.$$

Effective Potential at Electroweak Scale

- At electroweak temperatures, the Higgs background is described by the finite-temperature effective potential,

$$
\begin{aligned}
V_{\text{eff}}(\Phi, T) &= V_0(\Phi) + V_1(\Phi) + V_1^T(\Phi, T), \\
V_0(\Phi) &= -\frac{m^2}{2}\Phi^2 + \frac{\lambda}{4}\Phi^4, \\
V_1(\Phi) &= \sum_i n_i(-1)^{2s_i}\frac{1}{64\pi^2}m_i^4(\Phi)\left[\ln\left(\frac{m_i^2(\Phi)}{\mu^2}\right) - C_i\right], \\
V_1^T(\Phi, T) &= \sum_{i=\text{boson}} n_i\frac{T^4}{2\pi^2}J_b\left(\frac{m_i^2}{T^2}\right) - \sum_{j=\text{fermion}} n_j\frac{T^4}{2\pi^2}J_f\left(\frac{m_j^2}{T^2}\right).
\end{aligned}
$$

- Here i represents the particle, s_i its spin, n_i the number of degrees of freedom associated to the specific particle, μ is the renormalization scale and C_i are constants depending on the renormalization scheme. For example $n_g = 3$ for the gauge bosons, $n_q = 12$ for the quarks and $n_l = 4$ for the leptons.

Numerical Inputs

Electroweak-scale choices

$$T \approx 100\,\mathrm{GeV}, \qquad \mu = m_t = 173.3\,\mathrm{GeV},$$
$$g_* = 106.75.$$

- Higgs quartic coupling at tree level:

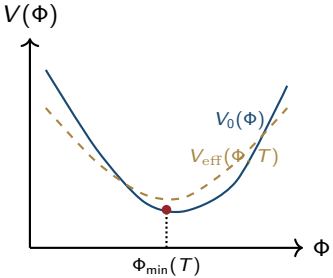
$$\lambda = \frac{m_h^2}{2v^2} \approx 0.1291.$$

- With one-loop thermal corrections:

$$\lambda(T) \approx 0.1317.$$

- Thermal minimum of the effective potential:

$$\Phi_{\min}(T = 100\,\mathrm{GeV}) \approx 216.6\,\mathrm{GeV}.$$



Solid curve: $V_0(\Phi)$

Dashed curve: $V_{\mathrm{eff}}(\Phi, T)$ at $T \approx 100\,\mathrm{GeV}$.

Numerical Estimations of Baryon to Entropy Ratio

- The net baryon density from the calculation is

$$\eta_1 = \frac{\Delta_B B}{g_*} = \frac{24i\,\delta(0)_{\rm reg}}{g_*\,T^4} \times \sin(\arg_b) \times \exp[-\Gamma_{\rm eff}(A)]\,.$$

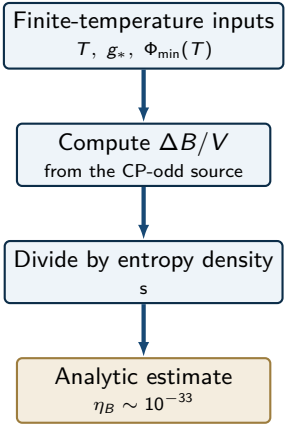
- Inserting the thermal minimum and the CP-odd source gives the analytic Standard Model estimate

$$\eta_B^{\rm analytic} \approx 10^{-33}.$$

- The η_1 is related to $\Delta_B B$ as:

$$\eta_1 = \frac{n_B}{s}, \qquad s \approx g_* n_\gamma, \qquad \eta_1 = \frac{n_B}{g_* n_\gamma}.$$

$$\boxed{\eta_1 \approx \frac{\Delta_B B}{g_*}}$$



Numerical Result: The Standard Model Falls Short

Outcome

$$\eta_1^{\text{analytic}} \equiv \frac{n_B}{s} \approx 10^{-33}$$

computed numerically.

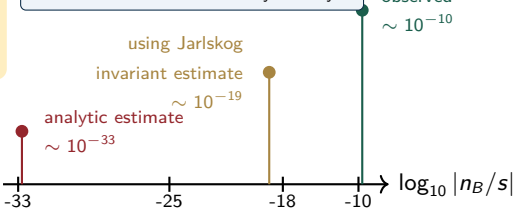
- Jarlskog invariant :

$$\begin{aligned} J &\approx c_{12}s_{12}s_{23}s_{13}\sin\delta \\ &\approx (0.9745)(0.2245)(0.0410)(0.0037)(0.93) \\ &\approx 3.0\times 10^{-5}. \end{aligned}$$

- Dimensional Estimation:

$$\delta_{CP} \approx J\,\frac{(m_t^2-m_c^2)(m_t^2-m_u^2)(m_c^2-m_u^2)}{T_{EW}^{12}} \times (m_b^2-m_s^2)(m_b^2-m_d^2)(m_s^2-m_d^2) \approx 10^{-19}.$$

The full flavour contraction shows the Standard Model result remains far below the observed asymmetry.



Conclusions

- Electroweak theory naturally permits baryon-number violation through anomaly-driven changes in the Chern–Simons number.
- The analysis translates CKM CP violation into the effective action by rotating the down-quark mass matrix, making the CP-odd source explicit.
- At the electroweak scale, the finite-temperature calculation gives a baryon asymmetry far below the observed value.
- The central physics message is therefore strong: Standard Model electroweak baryogenesis is quantitatively insufficient.

Final take-away

$\eta_B^{\mathrm{SM}} \ll \eta_B^{\mathrm{obs}}$, so additional CP violation and/or new electroweak dynamics are required.

Future Perspectives And BSM

What this analysis establishes

- Baryon-number violation exists.
- The CP source can be computed systematically.
- Thermal effects can be included consistently.

What still fails in the Standard Model

- CKM CP violation is too weak.
- The electroweak transition is not strongly first-order for $m_h \approx 125\,\mathrm{GeV}$.
- The predicted asymmetry remains many orders below observation.

Beyond-SM CP phases → Explains Sakharov Conditions → Successful baryogenesis

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Q&A

Thank You!

Questions and discussion are welcome!